



National Institute of Fashion Technology
Bhubaneswar

Department of Fashion Technology

Assignment –2

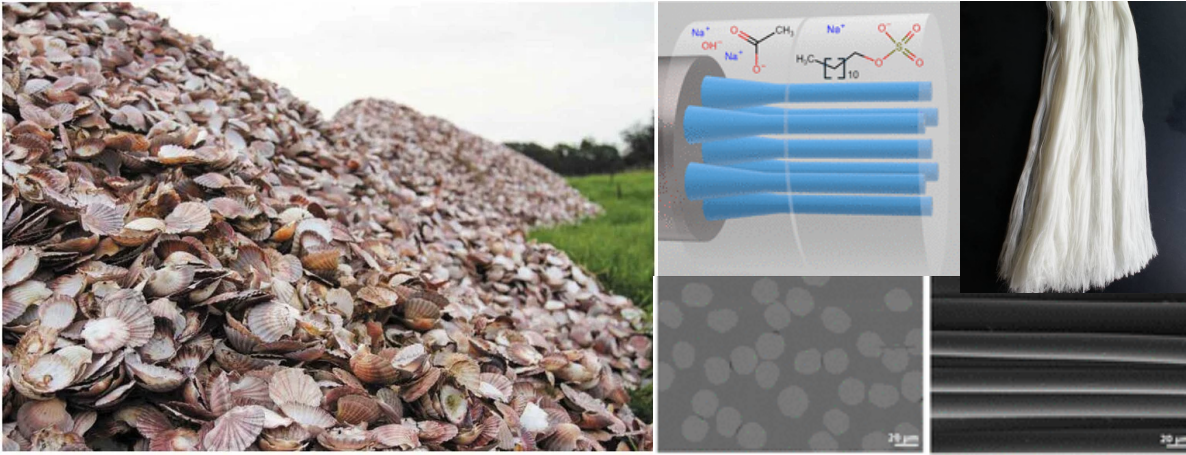
Subject:Sustainable Production

Submitted by: Shivadharshini R.A

Roll No:BFT/23/777

Submitted to: Dr.Sulagana Saha, Asst.Prof.

Date:13/04/2026



The Chitosan Fibre:

Article about its Composition, Processing and its Sustainability role

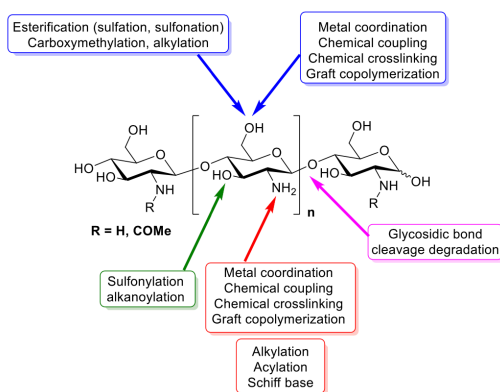


Fig 1 Fig 1 shows the chemical structure of chitosan and its reactive functional groups enabling various modifications.

While most sustainable fibres are derived from plants here comes fibre that is derived from largest phylum of the animal kingdom "Chitosan fibre", they are derived from chitin material which are obtained from the hard outer shell(exoskeleton)which are rich in chitin present in crustacean and few Insects and fungi.

It is a natural polymer made up of repeating sugar-based units arranged in a long chain structure(polysaccharide), containing functional groups(-NH₂, -OH) these are the major factors that contributes for its antibacterial property and support absorption . Its moisture absorbing ability further enhance its sustainability for textile use .

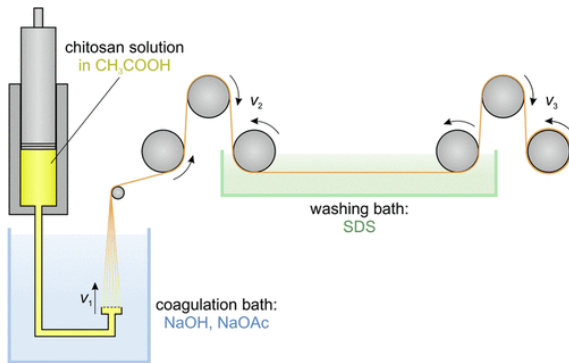


Fig 2 The figure illustrates the wet spinning process used for the formation of chitosan fibres.

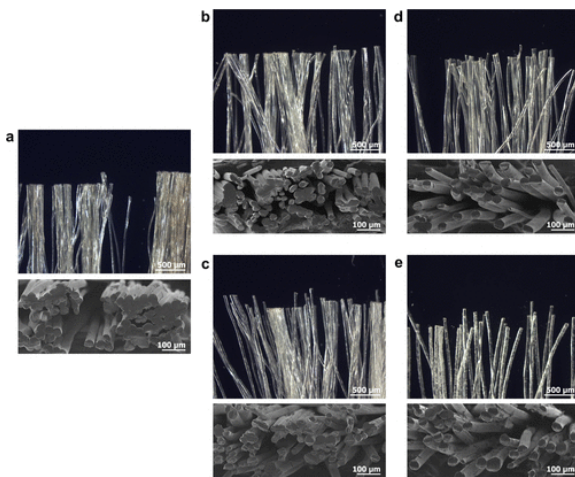


Fig 3: The images show the morphology of chitosan fibres, highlighting their structure and surface characteristics at different scales.

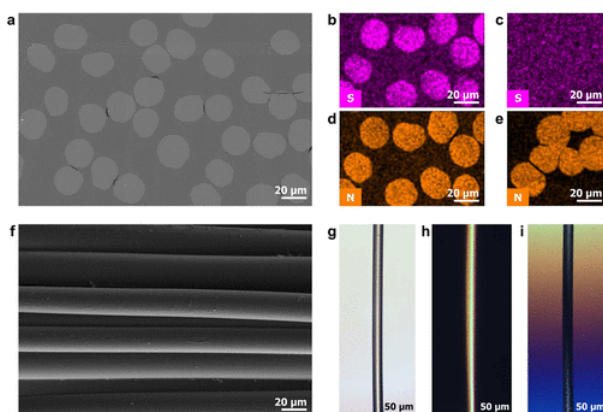
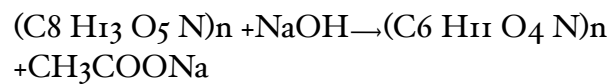


Fig 4 The figure presents the micro structural and compositional characteristics of chitosan fibres at microscopic levels

Now let's know about how the chitin is converted into a fibre (Chitosan). The initial step is to collect the shells which are rich in chitin a base material (basically found in the seafood waste industry and few from insects and fungi). The raw material (i.e) the shells also contain other unwanted elements (minerals, protein and pigments) which are not used for production of Chitosan. To remove these waste and to isolate chitin, the raw material undergo a process called cleaning and crushing (deproteinisation and demineralisation) and with this step pretreatment process is completed.

The next step is the key process where the chitin is converted into chitosan by the removal of Acetyl groups ($-\text{COCH}_3$) this is called deacetylation



(The above equation says when the deproteinised and demineralised chitin is reacted with NaOH (alkali) and gives Chitosan and Sodium acetate as by product)

Fibre formation :

The chitosan is dissolved in mild acidic solution because it is not easily soluble in water but acid makes a viscous solution (dope). This dope solution is passed through a spinneret and leads to the formation of thin continuous filaments, the obtained filaments are entered into a coagulation bath which contains alkaline solution that is further allowed to solidify and take shape for its alignment of the molecular structure.

The modified fibres undergoes drawing and stretching process that improves the strength and durability. The fibre is washed so the chemical are removed and neutralised by balancing the PH of the fibre.

The production of the Chitosan fibre is a significant choice for sustainable development within textile industry derived from crustacean shells waste by utilising the by product from the seafood industry . The process used for the production of(wet gel spinning) consume less energy comparatively to other process(melt spinning ,or solution spinning) which contain more chemical energy and also need more chemical recovery , the use of mild acidic solution and sodium acetate(CH_3COONa) released from the process of deacetylation can be indirectly used in textile industry for dyeing process to control PH and reused in chemical industry as a buffering solution that can reduce the chemical waste ,It can be used to make heat pack a (phase change material) that further enhance sustainability . Additionally the antibacterial property in chiton which is naturally present in the structure avoids the usage of additional chemical to attain this property that ensure the eco friendly production in the growth to textile industry.

On considering to its limitation ,it is not suitable for large scale application. One of the major challenge is these fibre have low mechanical strength in wet state (as per the study from Morphology and mechanical properties of chitosan)

The influence of different molecular weights of chitosan ,its degree of acetylation and fibre post treatment is essential to improve its mechanical strength. That impact in wide range of application such as odour resistant textile ,in sportswear, medical dressing with healing and bacteriostatic properties, food storage products with fungistatic and antibacterial effect as well as product for water purification .

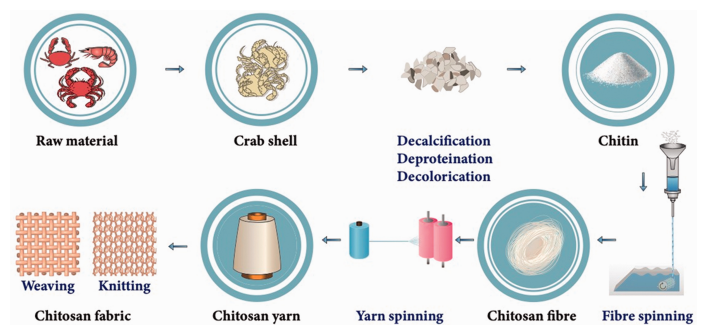


Fig 5: The figure illustrates the overall process flow of chitosan fibre production from raw crustacean shells to final textile applications

Reference :

1. Notin, L., Viton, C., David, L., Alcouffe, P., Rochas, C., & Domard, A. (2006). *Morphology and mechanical properties of chitosan fibers obtained by gel-spinning: Influence of the dry-jet-stretching step and ageing. Carbohydrate Polymers*
2. Rinaudo, M. (2006). *Chitin and chitosan: Properties and applications. Progress in Polymer Science, 31(7), 603–632*

Image source :

1. *Application of chitosan in the form of textile: production and sourcing*
2. Notin, L., Viton, C., David, L., Alcouffe, P., Rochas, C., & Domard, A. (2006). *Morphology and mechanical properties of chitosan fibers obtained by gel-spinning: Influence of the dry-jet-stretching step and ageing. Carbohydrate Polymers*
3. *Wet Spinning of Chitosan Fibers: Effect of Sodium Dodecyl Sulfate Adsorption and Enhanced Dope Temperature*